

BHAVIKA DEVNANI

bhavika.devnani@gmail.com • <https://bdevnani3.github.io/>

Education

Georgia Institute of Technology

PhD in Computer Science

Aug 2024 – Present

Atlanta, Georgia

- PhD Candidate in Hoffman AI Research Lab, advised by Judy Hoffman and James Hays.

Georgia Institute of Technology

Masters in Computer Science & Neuroscience

Jan 2021 – Dec 2022

Atlanta, Georgia

- Concentration in Machine Learning, GPA: 4.0
- Advised by Judy Hoffman and Dhruv Batra.

Georgia Institute of Technology

Bachelors in Computer Science

Aug 2013 – Aug 2017

Atlanta, Georgia

- Concentration in Artificial Intelligence and Theory

Publications

Training-free Adaptive Frame Selection for Video-Language Understanding

Bhavika Suresh Devnani, Jitesh Jain, Humphrey Shi, Judy Hoffman

Under Review at ICLR 2025

ELSA: Learning Spatially Aware Language-Audio Embeddings

Bhavika Suresh Devnani, Skyler Seto, Zak Aldeneh, Alessandro Toso, Elena Menyaylenko, Jonathan Sheaffer, Miguel Sarabia

Accepted at NeurIPS 2024 - 24% Acceptance Rate - Tier 1 Conference

ZSON: Zero-Shot Object-Goal Navigation using Multimodal Goal Embedding

Arjun Majumdar, Gunjan Aggarwal*, Bhavika Suresh Devnani, Judy Hoffman, Dhruv Batra*

Accepted at NeurIPS 2022 and CoRL 2022 Workshop

Bi-Directional Self-Attention for Vision Transformers

George Stoica, Taylor Hearn, Bhavika Suresh Devnani, Judy Hoffman

Accepted at NeurIPS Vision Transformers Workshop 2022 – **Won Best Paper**

Experience

Apple (AIML) – ML Research Intern

Cambridge, UK

May 2025 – Aug 2025

- Work on AIML team led by Dr. Samy Bengio
- Explore methods for improving data efficiency of Transformer-based architectures, specifically exploring higher order formulations of attention.

Apple (AIML) – ML Research Engineer

Cupertino, California

Dec 2022 – Aug 2024

- Work on AIML team led by Dr. Samy Bengio
- Optimize representation learning algorithms
- Focus: Multimodal foundation models, with emphasis on audio and language

Delivery Hero (Talabat) – Senior Data Scientist **Sept 2020 – Dec 2020**
Dubai, UAE

- Develop personalization models for restaurant recommendations and ML infrastructure pipeline
- Analyze vendor menu approval pipeline and develop automation plan with demo implementations
- Lead bi-weekly ML paper reading group

New York University – Syllabus Design **July 2020 – Aug 2020**
Abu Dhabi, UAE

- Design and write content for new course: Programming for Economists
- Create interactive lectures in Google Colab to encourage student engagement

Quora Inc – Machine Learning Engineer **Sept 2017 – March 2020**
Mountain View, California

- Improve accuracy of neural network for answer prediction, increasing revenue by 4%
- Migrate Machine Learning Model Service to Kubernetes
- Engineer feature logging pipelines ensuring consistent features across datastores
- Improve "Ask to Answer" service stability and performance by ~2%
- Architect and build in-house logging library for C++ services

LinkedIn (Microsoft) – Software Engineering Intern **May 2016 – Aug 2016**
Mountain View, California

- Develop security tool that predicts spam invitation probability
- Design and present tool specifications; complete project from conception to production

Pindrop Security – Software Engineering Intern **May 2015 – Aug 2015**
Atlanta, Georgia

- Refactor and develop web scrapers for data collection – deployed to production
- Create Android application using Pindrop API to block and report spam calls

Teaching

Georgia Institute of Technology – Teaching Assistant **Jan 2014 – Current (With Breaks)**
Graduate Deep Learning (Fall '21, Spring '22) • Graduate Machine Learning (Spring '21, Summer '22) • Introduction to Computer Science (Spring '14, Fall '14, Summer '17) • Discrete Math - Honors (Spring '17, Fall '15) • Algorithms (Spring '16, Fall '16) • Automata and Complexity (Spring '15)

Service & Volunteering

Conference Reviewer – Online English Tutor **Summer 2021 – Present**
Acted as a peer review for multiple top conferences in the domain of Artificial Intelligence: NeurIPS '22, ICLR '22, NeurIPS '23, ICLR '23 NeurIPS '24, NeurIPS '25

StreetCode – Mentor **Jan 2018 – March 2020**
Teach computer science to students from underserved communities in East Palo Alto

Georgia Institute of Technology – RiseUp Tutor & Researcher **Jan 2016 – Aug 2017**
Tutor high school students in AP Java; analyze learning patterns from textbook interaction logs